Neeraj Singh Guleria

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SUMMARY

Data Engineer with 2+ years of experience designing and deploying scalable ETL pipelines, distributed data processing workflows, and business intelligence dashboards on Microsoft Azure. Proven track record automating 1 million+ monthly production logs and reducing analysis time by 90%. Proficient in PySpark, Python, SQL, Azure Data Factory, Azure Databricks, Azure Synapse Analytics, Delta Lake, and Snowflake. Hands-on experience with medallion architecture (Bronze, Silver, Gold), star schema data modeling, SCD Type 2, incremental loading, and real-time streaming. Passionate about building reliable, governed, and scalable data infrastructure that drives operational decisions.

WORK EXPERIENCE

Data Engineer – **Optiemus Electronics Ltd.**, Noida, India *March 2024 – Present*

- Designed and deployed end-to-end ETL pipelines using Azure Data Factory (ADF) to ingest data from MES, ERP, SMT machines, and on-premises Oracle systems into Azure Data Lake Storage Gen2.
- Processed and transformed 1 million+ monthly production test logs using Azure Databricks and PySpark; stored outputs as optimized Delta Lake tables with Z-ordering and auto-compaction.
- Implemented Bronze, Silver, and Gold medallion architecture layers supporting defect tracking, yield analysis, and OEE (Overall Equipment Effectiveness) reporting across manufacturing lines.
- Built incremental data ingestion pipelines using watermarking strategies and ADF tumbling windows, significantly reducing redundant reprocessing and pipeline runtime.
- Designed Synapse SQL dedicated pool schemas with Fact Production, Fact Defects, Dim Machine, and Dim Product tables following star schema data modelling best practices.
- Automated station-level performance reporting for SMT, Final Assembly, and Packaging lines — saving 100+ engineering hours per month and eliminating manual analysis bottlenecks.
- Secured all pipeline credentials and connection strings using Azure Key Vault integration across ADF linked services.
- Configured real-time pipeline monitoring, alerting, and failure notifications using Azure Monitor and Log Analytics workspaces.
- Contributed to Unity Catalog implementation in Azure Databricks for data governance, role-based access control (RBAC), and metadata management.
- Developed metadata-driven pipeline framework under senior guidance to improve scalability, reusability, and maintainability across 20+ ADF pipelines.
- Wrote advanced PL/SQL stored procedures for optimized MES data extraction, reducing query execution time for downstream pipeline ingestion.
- Built and maintained automated Power BI dashboards (Yield, OEE, Defect Codes, Cycle Time) used daily by production heads and senior management for data-driven decision making.

Data Analyst Intern – **Living Things**, Mumbai, India *October 2023 – January 2024*

- Monitored and analyzed key business datasets using Python and SQL to surface actionable insights, supporting improved operational efficiency and strategic planning.
- Built Excel-based reporting models and automated recurring data validation and transformation workflows, reducing manual reporting effort.
- Created data visualizations and summaries to communicate findings clearly to non-technical stakeholders.

Network Intern – **Ekaga Group**, Delhi, India *May 2022 – June 2022*

- Configured and tested a GPON network for an IPMPLS Defense Project, supporting IP address planning, hardware setup, and network troubleshooting.
- Gained hands-on exposure to network operations, TCP/IP protocols, and network security fundamentals.

SKILLS

Cloud Platforms: Microsoft Azure, Azure Data Factory (ADF), Azure Databricks, Azure Synapse Analytics, Azure Data Lake Storage Gen2 (ADLS Gen2), Azure Event Hubs, Azure Cosmos DB, Azure SQL Database, Azure Key Vault, Azure Purview, Snowflake

Big Data & Processing: Apache Spark, PySpark, Spark Structured Streaming, Apache Kafka, Delta Lake, Apache Airflow

ETL & Data Engineering: dbt (Data Build tool), ETL/ELT Pipeline Design, Medallion Architecture, Incremental Loading, Watermarking, SCD Type 2, Metadata-Driven Pipelines, Data Vault 2.0, REST API Integration

Databases & Warehousing: SQL (Advanced), PL/SQL, Azure Synapse Dedicated SQL Pools, Oracle, MySQL, Star Schema, Snowflake Schema

Programming: Python, PySpark, Pandas, NumPy, SQL, PowerShell, Bash, C#, JavaScript

Data Modelling & Governance: Star Schema, Snowflake Schema, Data Vault 2.0, RBAC, Data Lineage, Unity Catalog, Azure Purview

DevOps & CI/CD: GitHub Actions, Terraform, Azure DevOps, ARM Templates, Bicep, Docker

BI & Monitoring: Power BI, Azure Monitor, Log Analytics

Data Formats: Parquet, Avro, JSON, Delta Lake

PROJECTS

Azure Manufacturing Data Pipeline

- Problem: Manual production log analysis took several hours per cycle, creating delays in defect detection and yield reporting for 1 million+ monthly records.
- Solution: Built a complete ETL pipeline using Azure Data Factory, Azure Databricks (PySpark), and Azure Synapse Analytics with medallion architecture, Delta Lake optimization, data validation checks, and Azure Monitor alerting.
- Outcome: Reduced analysis time by 90%, enabled real-time Power BI dashboards, and fully automated reporting — recovering 100+ engineering hours per month.

MES Web Application for Self-Serve Production Reporting

- Problem: Production teams depended on IT to run complex SQL queries for historical data, creating bottlenecks and delays in operational decisions.
- Solution: Developed an ASP.NET Core MVC web application with dynamic filters, server-side pagination, and optimized PL/SQL stored procedures for fast data retrieval.
- Outcome: Reduced manual query turnaround time by 80%, enabling production teams to self-serve real-time reports without IT involvement.

End-to-End Azure Data Engineering Pipeline

- Problem: Organization lacked scalable data infrastructure for business intelligence reporting and executive KPI tracking.
- Solution: Designed and implemented full medallion architecture using ADF, Azure Databricks, Synapse Analytics, and Power BI, including incremental loads, SCD Type 2 handling, and star schema modelling.
- Outcome: Delivered 3 executive dashboards backed by 5 fact tables and 12 dimension tables, improving data accuracy, consistency, and reporting speed across teams.

EDUCATION

Bachelor of Science (Hons.) – Data Science *2020 – 2023*

K.R. Mangalam University, Gurgaon, Haryana

Relevant coursework: Statistical Data Analysis, Python, SQL, Data Mining, Machine Learning, Power BI, Database Management Systems (DBMS)

CERTIFICATIONS

- Azure Data Engineering Masters: Build Scalable Solutions – Udemy (2025)
- Azure Data Engineering for Data Engineers: DP-203 and Beyond – Udemy (2026)